Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/11/2006
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Project Title:	Polyarylene Ether and Polymide Materials with Improved Radiation Shielding Capabilities		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23187-8795	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1900		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To develop novel, high-strength plastics with chemical compositions which have been modified from those in use today so that they have the added benefit of reducing the effects of cosmic radiation in the interior of spacecraft and lunar and Martian habitats. Collisions between the high-energy (HZE) particles in cosmic radiation with atoms in the metal shell of a spacecraft result in nuclear fission which can lead to dangerous doses of radiation inside of a spacecraft. However, materials having high numbers of hydrogen atoms per unit mass are known to be effective shields. In this task, high-strength plastic materials are being formulated and prepared with large quantities of hydrogen. For this work , aromatic polyimides and polyarylene ethers were selected because of their thermal stability and mechanical strength. To provide light-weight polyimide and polyarylene ether materials for (i) shielding from cosmic radiation inside spacecraft and extraterrestrial habitats while (ii) also serving other functions possibly including structural support, thermal and acoustic insulation, micrometeorite shielding, plumbing, and space suits.		

	Plastic components in the living and working quarters of spacecraft will add to the protection of both humans and electronic devices while simultaneously serving as part of the spacecraft's structure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential materials that might provide radiation shielding for high-flying aircraft.
Task Progress:	Background: Long-duration manned travel beyond the van Allen radiation belts is impossible without adequate shielding from the high-energy, high-charge (HZE) particles in GCR. Shields that would be practical are not available. With current shielding capabilities, manned missions beyond the van Allen belts are limited to brief forays such as the Apollo missions to the moon. The high velocity HZE nuclei in GCR can be slowed and eventually stopped in a shield when the projectile nuclei interact Coulombically with the electrons and nuclei in the shield and when, with less frequency, the fragmentation occurs as the result of a direct collision of the projectile nucleus with a shielding nucleus. Hydrogen is a strong candidate as a component of a GCR shield. It is the only element whose atoms have no neutrons. Consequently, hydrogen's nucleus cannot fragment and both its component particles (electrons and protons) carry charge for Coulombic interactions. Consequently hydrogen is the most effective atomic species for shielding on a unit mass basis. Goal: We proposed to develop lightweight polymeric materials that could serve as radiation shielding and also serve ancillary functions such as in mechanical structures, insulation, and micrometeorite shielding. Some polymers such as polyethylene, designated "aliphatic," have a chemical composition with a large proportion of hydrogen atoms, rendering these polymers good candidates as GCR shields. Other polymers such as Kapton are "aromatic" polymers. Their chemical structures contain a small proportion of hydrogen atoms and, therefore, aromatic polymers are less desirable for GCR shielding. On the other hand, aromatic polymeric materials, as a class, tend to have superior mechanical and thermal properties in comparison to aliphatic polymeric materials. The goal in the studies supported by NAG8-1900 was to synthesize and characterize hybrid polymers that (i) are lightweight, (ii) have more desirable mechanical properties than purely aliphatic polymers have and (iii) also offer greater shielding capability against galactic cosmic radiation (GCR) than highly aromatic polymers. Novel Candidate Polymers for GCR Shielding: Toward this goal, we have prepared aromatic polymers–polyimides and polyethers–with chemical structures containing "extra" hydrogen atoms usually in the form of methyl groups. Specifically we have • synthesized and characterized three new polyimdes whose chemical structures are based on hydrogen-rich 1,2,3,4-butanetetracarboxylic dianhydride, to yield polymers having hydrogen concentrations of 0.052–0.062 moles/gram; • synthesized a new polyimide based on the hydrogen-rich diamine 1,4-diaminocyclohexane; • synthesized two new hydrogen-rich dianhydrides (designated "TMA" and "BTMA") and prepared two novel polyimides from them • further characterized novel aromatic polyimides and polyethers whose syntheses were carried out as part of this project, but prior to the period covered by this report. Application in a Graphite-Polymide Composite: The aromatic polyimide prepared from 4,4'-(4,4'-isopropylidenediphenyl-1,1'-diyldioxy)-dianiline and 4,4'-(4,4'-isopropylidenediphenoxy)bis(phthalic anhydride) was coated on carbon fiber. (This polyimide was first synthesized prior to the period covered by this report.) Pieces of the resulting prepreg tape were stacked and pressed at elevated temperatures. Scanning electron microscopy and ultrasonics revealed voids in the structure. The grant expired before further studies could be undertaken. Computer Simulations: We carried out computer modeling of the radiation shielding materials in the space radiation environment. This was undertaken in an effort to understand radiation testing performed at the Brookhaven National Laboratory (BNL) on our samples. Transport codes HZETRN (simulates the full spectrum of particles and energies in GCR) and GRNTRN (simulates the interactions of a single isotopic nucleus at a single energy as observed at BNL), both of which were developed at NASA-Langley, were employed.. In addition calculations were also performed with the code GEANT4 as developed at CERN. Comparisons of the results frm the different codes were partially successful. Publication: The manuscript "Aliphatic/Aromatic Hybrid Polymers for Functionally Graded Radiation Shielding" has been published: L. Y. Hu, S. Yang, A. K. Miller, C. S. Park, K. A. Plichta, S. J. Rochford, M.E. Schulz, R. A. Orwoll, B. J. Jensen, High Performance Polymers 18, 213-225 (2006).
Bibliography Type:	Description: (Last Updated: 03/30/2017)
Articles in Peer-reviewed Journals	Hu LY, Yang S, Miller AK, Park CS, Plichta KA, Rochford SJ, Schulz ME, Orwoll RA, Jensen BJ. "Aliphatic/Aromatic Hybrid Polymers for Functionally Graded Radiation Shielding." High Performance Polymers. 2006 Apr;18(2):213-25. , Apr-2006